

Brian E. Ramirez Zea

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Objective

Aspiring software engineer in DevOps and backend development, seeking to leverage strong programming skills, cloud infrastructure knowledge, and a passion for building scalable systems in a dynamic software engineering role.

Education

University of Puerto Rico, Mayagüez, B.S. in Software Engineering Aug 2021 – May 2028

- Major GPA: 3.05/4.00
 - Relevant Coursework: Data Structures, Operating Systems, Software Engineering, Computer Architecture, Algorithms & System Design
- CodePath Tech Exchange**, Certificate in Applied Software Development & Systems Jan 2026 – May 2026

- Selected as an Immersive Software Engineering Fellow for an intensive 15-week academic program in partnership with UPRM.
 - Core Modules: Advanced Software Engineering with GenAI, System Design & Architecture, Data Structures & Algorithms, Careers in Tech.
- CodeKloud**, Certificate in CKA (Certified Kubernetes Administrator) June 2026 – present

- completed 50% of the CKA course and hands-on labs, focusing on Kubernetes cluster management, networking, and security.
- Core Modules: Kubernetes Architecture, Cluster Setup & Configuration, Workloads & Scheduling, Services & Networking, Storage, Logging & Monitoring, Security, Troubleshooting.

Skills

Languages: Python (Advanced), Java, C++ , JavaScript, TypeScript, SQL (PostgreSQL), MIPS/6502 Assembly, Bash/Shell

Systems & Cloud: Linux (Arch, Ubuntu/Debian), Docker, Kubernetes (K3s), Oracle Cloud (OCI), Google Cloud (Cloud Run, BigQuery, Vertex AI), Raspberry Pi 5, Tailscale, Cloudflare Tunnels

Frameworks & Tools: FastAPI, Flask, React, Angular 19, Node.js, REST APIs, GitHub Actions (CI/CD), Gitflow, PostgreSQL, OpenAI/Gemini APIs, Systemd, UFW

Experience

Administrative Assistant, University of Puerto Rico, Mayagüez – Mayagüez, PR Aug 2026 – present

- Streamlined departmental operations and supported administrative workflows to ensure prompt and accurate institutional service delivery.
- Managed front-desk communications, schedule coordination, and student record management while providing direct procedural assistance to faculty and students.

Advanced Programming Mentor, University of Puerto Rico, Mayagüez – Mayagüez, PR Jan 2025 – May 2025

- Elevated student comprehension and success through one-on-one technical troubleshooting and guidance on complex programming assignments.
- Strengthened mentees' problem-solving velocity and code quality by reinforcing core concepts in memory management, recursion, and object-oriented design.

Projects

UnifiedVault (Hybrid Cloud Homelab & Monorepo) 2025 – Present

Multi-node hybrid private cloud orchestration and zero-trust infrastructure

- Engineered a hybrid-cloud cluster bridging an edge Raspberry Pi 5 with Oracle Cloud Infrastructure (OCI) instances, enabling scalable distributed processing.
- Hardened system security to a zero-trust standard using Tailscale mesh VPN and UFW firewall rules, eliminating public-facing port exposure for internal services.

StockSense (Backend Inventory Management System) 2026-02 – 2026-05

Transaction-based inventory management REST API and CLI tool

- Architected a centralized backend service using a transaction-based model to ensure data accuracy, traceability, and a verifiable audit trail for inventory tracking.
- Developed a lightweight REST API and CLI-based system featuring automated reordering logic to identify low-stock items based on historical usage patterns.

AirBets (Full-Stack Cloud Application) 2026-01 – 2026-05

Containerized full-stack web application integrating BigQuery and Vertex AI

- Designed and deployed a full-stack web application from the ground up using Python and Flask, integrating Generative AI capabilities into user workflows.
- Built a scalable backend utilizing Google Cloud BigQuery for data management and integrating Vertex AI APIs for data-driven generative features.